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GLM-5.1 AI | ARTIFICIAL INTELLIGENCE | FREE OPEN SOURCE MODEL

GLM-5.1 Just Beat GPT-5.4 and Claude Opus (And It's Free and Open-Source)

China's Free AI Just Topped The Leaderboard



Ai studio

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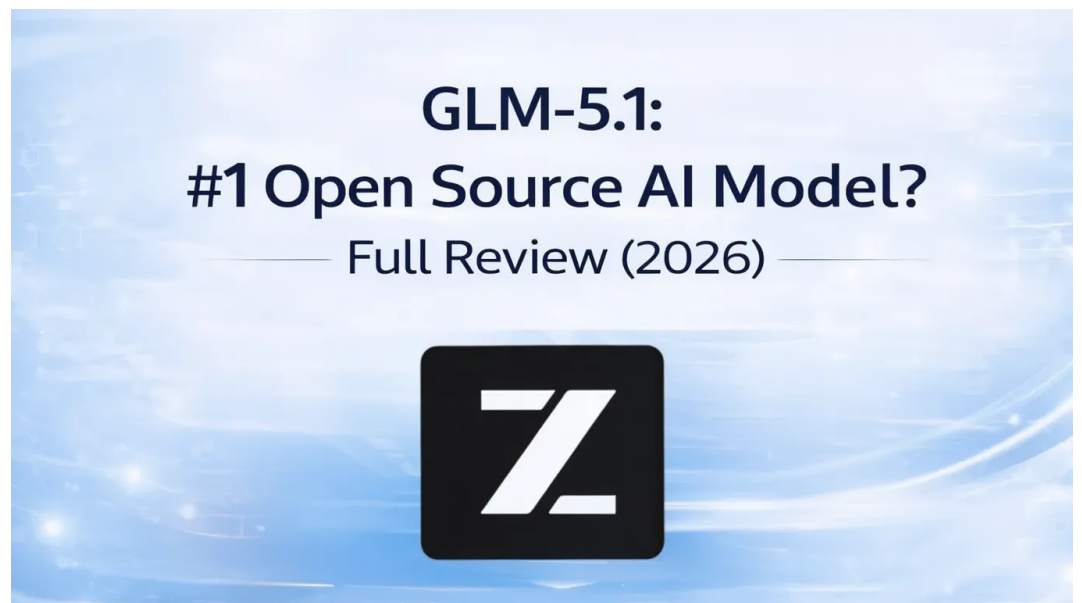


Image of GLM-5.1 (Image edited by Author)

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A free AI model from China just outscored GPT-5.4 and Claude Opus 4.6 on one of the most trusted coding tests in the field. It was released on April 7, 2026, its code is publicly available for anyone to download, and the company behind it built the whole thing without a single American chip.

That combination, top-tier performance, fully open, and built on entirely domestic Chinese hardware, is what's making people pay attention.

Who Made It and Why It Matters

The model is called GLM-5.1. It was built by Z.ai, a Chinese AI lab formerly known as Zhipu AI, which spun out of Tsinghua University. In January 2026, the company went public on the Hong Kong Stock Exchange, raising about \$558 million, becoming the first publicly traded AI foundation model company in the world.

That money shows.

Between February and April 2026, Z.ai released four significant model updates in under eight weeks. GLM-5.1 is the latest.

The model is free and open-source under the MIT license, which is about as permissive as a license gets. Anyone can download it, modify it, build a product with it, or sell something made from it. No fees, no restrictions, just a requirement to keep the copyright notice.

What Is SWE-Bench Pro, and Why Does the Score Matter?

Most AI benchmarks test things like trivia, math problems, or reading comprehension.

SWE-Bench Pro is different.

It takes real, unsolved bugs from real software projects on GitHub and asks the AI to fix them, on its own, without a human walking it through anything.

It is considered one of the most honest tests of whether an AI can actually do useful software work, not just sound impressive in a chat window.

On this test, GLM-5.1 scored 58.4.

For comparison:

- GPT-5.4 scored 57.7
- Claude Opus 4.6 scored 57.3
- Gemini 3.1 Pro scored 54.2

GLM-5.1 sits at the top. And it does it as a free, openly available model.

Working for 8 Hours Straight

Most AI tools work best when you ask them something clear and contained. Give them a short task, get an answer, move on. Ask them to manage a long, complex, multi-day project on their own, and they tend to fall apart. They lose track of what they were doing, repeat mistakes, or give up and declare the job done when it isn't.

GLM-5.1 was specifically built to avoid that problem.

In one test, the model was given a software optimization task and left to work on its own for eight hours. It ran over 655 rounds of self-testing, figured out where things were breaking down, changed its approach when something stopped working, and ended up making the system run nearly seven times faster than when it started.

In another test, it built an entire functional desktop environment from scratch. Not a mockup. A working file browser, a terminal, a text editor, a system monitor, and playable games, all built autonomously, all polished iteratively until they looked and worked consistently.

The Z.ai team described the model's behavior as a staircase pattern.

Instead of grinding at a problem until it hits a wall, the model recognizes when its current approach has run out of steam, shifts strategy, and opens up a new level of performance. Most AI tools run out of useful ideas after 20 or 30 steps. GLM-5.1 has been tested sustaining productive work across 1,700 steps.

The Hardware Story Is Quietly Significant

Since January 2025, Zhipu AI has been on the US Entity List. That is the US government's list of foreign companies that American businesses are restricted from trading with. In practice, **it means Z.ai cannot legally buy Nvidia GPUs, the hardware that almost every major AI lab in the world depends on for training their models.**

So Z.ai trained GLM-5.1 on approximately 100,000 Huawei chips instead, using Huawei's own software stack. No American silicon involved at any stage.

The reason this matters is simple.

One of the main arguments behind US export controls on AI chips is that restricting access to cutting-edge hardware will slow down foreign competitors. GLM-5.1 is a direct piece of evidence about how well that argument holds. A model trained entirely on non-American, domestically produced hardware just topped a globally respected benchmark.

That does not settle the policy debate. But it is a data point that anyone following the AI landscape should be aware of.

What It Is Not Good At

Honest coverage means being clear about the limitations.

GLM-5.1 cannot process images.

If you need an AI that can look at a screenshot, analyze a diagram, or review a visual output, this model cannot do that. Claude Opus 4.6 can. That is a real difference for certain kinds of work.

It is also slow compared to other top models.

When generating text, it moves at roughly 44 tokens per second, which is noticeably sluggish for quick back-and-forth tasks. It is built for sustained, deep work, not fast answers.

Running it locally on your own machine is possible, but not easy.

The full model requires 1.65 terabytes of storage. A compressed version exists that brings it down to around 220 gigabytes, small enough to run on certain high-end Mac setups, but performance suffers with compression. Most people accessing this model will do so through an API or a platform, not by running it themselves.

On the broader combined coding score that averages across multiple benchmarks together, Claude Opus 4.6 still holds a slight lead over GLM-5.1. The SWE-Bench Pro result is real and meaningful, but calling GLM-5.1 the best coding AI across the board would be overstating it.

Initial Take: Impressive but Needs Verification

It's a really impressive model, but at the same time, something to stay cautious about.

Performance on complex, long-running tasks looks strong. For things like multi-file refactoring, backend architecture, and work that needs sustained planning, it clearly stands out and feels like one of the top open-source options right now.

At the same time, the benchmark score is self-reported and hasn't been independently verified yet, so that's still an open question. Earlier results from the same lab did end up holding up under external testing, so there's some credibility there, but nothing confirmed for this specific version so far.

There's also the usual concern about benchmark optimization. Big numbers like these always raise that possibility, and it'll only become clearer once more real-world testing and independent evaluations happen.

The Bigger Pattern Behind One Model

Open-source AI has been catching up to private, closed models for three years now.

In 2023, open-source models were roughly two years behind the frontier. In 2024, that gap closed to about one year. By 2025, six months. In April 2026, an open-source model is sitting at the top of one of the most respected coding benchmarks in the world.

GLM-5.1 is not a one-off. Chinese AI labs including DeepSeek, Alibaba's Qwen team, and Moonshot AI are producing competitive open models at a pace that is starting to reshape where developers look when they need capable AI without an ongoing subscription.

One industry report noted that roughly 80% of AI startups are now gravitating toward open-source Chinese models. Whether that number is precisely right or not, the directional shift is real.

How to Actually Use It

If you want to try GLM-5.1:

- The model weights are free to download from HuggingFace at [zai-org/GLM-5.1](https://huggingface.co/zai-org/GLM-5.1)
- The Z.ai API charges \$1.40 per million input tokens and \$4.40 per million output tokens, which is significantly cheaper than Claude Opus 4.6's pricing
- It works directly inside tools like Claude Code, Cursor, and Cline without special configuration
- For local setup, the Unsloth team has released a compressed version that works on high-memory Mac hardware, though it runs slowly

For most people, the API is the simplest starting point.

The Bottom Line

GLM-5.1 is a free, open-source AI model that currently holds the top score on one of the most trusted software coding benchmarks in the world.

It was built without any American hardware by a Chinese company that is legally barred from buying Nvidia GPUs. It is especially good at long, autonomous tasks where it keeps improving its own work over hours rather than giving up early.

It is not the best model at everything. It cannot see images. It is slow for quick tasks. The top benchmark score is self-reported.

But for the kind of complex, sustained coding work that used to require an expensive private model subscription, there is now a free alternative that can hold its own at the highest level. That is what is actually new here.

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